

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.015, beta.var.scale = 0.4\n')
```

```
## sdev = 0.015, beta.var.scale = 0.4
```

Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.5467  0.5911  0.6000  0.5988  0.6061  0.6322

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.5794  0.7181  0.6616  0.8061  0.9672
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.5628  0.6989  0.6433  0.7839  0.9372
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.01500 0.04500 0.07111 0.23763 0.26722 1.00000
```

```

Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02111 0.04597 0.07056 0.20878 0.20833 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01778 0.05278 0.13444 0.33930 0.64111 1.00000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.3317  0.3533  0.3606  0.3609  0.3689  0.4017

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.3700  0.5522  0.4981  0.6722  0.9456
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.3611  0.5367  0.4797  0.6467  0.8833
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01556 0.40056 0.49691 1.00000 1.00000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.007222 0.040000 0.323940 0.855000 1.000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.5711  1.0000  0.7635  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.1406  0.1644  0.1717  0.1722  0.1789  0.2256

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2572 0.4561 0.4108 0.5867 0.9111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2567 0.4533 0.4022 0.5744 0.8394
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.9383 1.0000 0.8377 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.001111 0.013333 0.293895 0.757083 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.6106 1.0000 0.7663 1.0000 1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.08667 0.11889 0.12667 0.12870 0.13667 0.21000

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1378 0.3500 0.3272 0.4933 0.9022
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1429	0.3536	0.3211	0.4867	0.8006

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6483	0.8761	0.7410	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.003333	0.298514	0.759722	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9628	1.0000	0.8683	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.07556	0.09333	0.10333	0.10519	0.11389	0.21444

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08556	0.29333	0.28270	0.44167	0.86833

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0900	0.2989	0.2790	0.4383	0.7956

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3172	0.6361	0.5856	0.9283	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8683	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2616	0.6307	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8233	1.0000	0.8386	1.0000	1.0000

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2078	0.2300	0.2411	0.2438	0.2544	0.3339

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02611	0.15222	0.19217	0.31333	0.85167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02778	0.14944	0.18017	0.30444	0.68833

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2897	0.5967	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7572	0.8361	0.8218	0.9000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1129	0.0000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2117	0.5444	0.5406	0.9817	1.0000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.05500	0.08056	0.09167	0.09543	0.10611	0.24278

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.008889 0.207222 0.218045 0.368889 0.800000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0100	0.2194	0.2178	0.3761	0.6928

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.3572	0.3992	0.7311	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.6711	1.0000	1.0000	0.9977	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1903	0.3443	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5411	0.7678	0.6626	0.8933	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2694	0.2972	0.3072	0.3103	0.3211	0.4450

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.009444	0.118333	0.175001	0.277222	0.799444

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01111 0.11389 0.13899 0.23222 0.61944
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.06554 0.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.5644  0.6889  0.6881  0.8122  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.0275  0.0000  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.1800  0.2844  0.5156  1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.1961  0.2239  0.2339  0.2396  0.2511  0.3661

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.003889 0.087778 0.154732 0.261111 0.766111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.087778 0.135438 0.245556 0.636667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000  0.000  0.000  0.177  0.335  1.000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.6700 0.7611 0.7722 0.8861 1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.06023 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```
0.00000 0.00000 0.07611 0.23007 0.43222 1.00000
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5467  0.5911  0.6000  0.5988  0.6061  0.6322
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5794  0.7181  0.6616  0.8061  0.9672
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5628  0.6989  0.6433  0.7839  0.9372
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01500 0.04500 0.07111 0.23763 0.26722 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02111 0.04597 0.07056 0.20878 0.20833 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01778 0.05278 0.13444 0.33930 0.64111 1.00000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3317  0.3533  0.3606  0.3609  0.3689  0.4017
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.0000 0.3700 0.5522 0.4981 0.6722 0.9456
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.3611 0.5367 0.4797 0.6467 0.8833
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.01556 0.40056 0.49691 1.00000 1.00000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.007222 0.040000 0.323940 0.855000 1.000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.5711 1.0000 0.7635 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.1406 0.1644 0.1717 0.1722 0.1789 0.2256
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.2572 0.4561 0.4108 0.5867 0.9111
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.2567 0.4533 0.4022 0.5744 0.8394
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9383 1.0000 0.8377 1.0000 1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.001111 0.013333 0.293895 0.757083 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.6106 1.0000 0.7663 1.0000 1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.08667 0.11889 0.12667 0.12870 0.13667 0.21000
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1378 0.3500 0.3272 0.4933 0.9022
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1429 0.3536 0.3211 0.4867 0.8006
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.6483 0.8761 0.7410 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.003333 0.298514 0.759722 1.000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9628 1.0000 0.8683 1.0000 1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.07556 0.09333 0.10333 0.10519 0.11389 0.21444
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.08556 0.29333 0.28270 0.44167 0.86833
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0900 0.2989 0.2790 0.4383 0.7956
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.3172 0.6361 0.5856 0.9283 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8683 1.0000 1.0000 1.0000 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2616 0.6307 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.8233 1.0000 0.8386 1.0000 1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2078 0.2300 0.2411 0.2438 0.2544 0.3339
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02611 0.15222 0.19217 0.31333 0.85167
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
## Modality 5 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
## 0.00000 0.02778 0.14944 0.18017 0.30444 0.68833
## Modality 2 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
## 0.0000  0.0000  0.0000  0.2897  0.5967  1.0000
## Modality 3 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
## 0.0000  0.7572  0.8361  0.8218  0.9000  1.0000
## Modality 4 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
## 0.0000  0.0000  0.0000  0.1129  0.0000  1.0000
## Modality 5 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
## 0.0000  0.2117  0.5444  0.5406  0.9817  1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
## 0.05500 0.08056 0.09167 0.09543 0.10611 0.24278
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
## 0.000000 0.008889 0.207222 0.218045 0.368889 0.800000
## Modality 2 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
## Modality 3 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
## Modality 4 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
## Modality 5 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
## 0.0000  0.0100  0.2194  0.2178  0.3761  0.6928
## Modality 2 :
## Min. 1st Qu. Median      Mean 3rd Qu.      Max.
## 0.0000  0.0000  0.3572  0.3992  0.7311  1.0000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6711  1.0000  1.0000  0.9977  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1903  0.3443  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5411  0.7678  0.6626  0.8933  1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2694  0.2972  0.3072  0.3103  0.3211  0.4450
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.009444 0.118333 0.175001 0.277222 0.799444
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01111 0.11389 0.13899 0.23222 0.61944
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.06554 0.00000 1.00000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5644  0.6889  0.6881  0.8122  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.0275  0.0000  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.1800  0.2844  0.5156  1.0000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.1961  0.2239  0.2339  0.2396  0.2511  0.3661
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000000 0.003889 0.087778 0.154732 0.261111 0.766111
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000000 0.004444 0.087778 0.135438 0.245556 0.636667
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.000 0.000 0.000 0.177 0.335 1.000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.0000 0.6700 0.7611 0.7722 0.8861 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.00000 0.00000 0.00000 0.06023 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.00000 0.00000 0.07611 0.23007 0.43222 1.00000
##
## ‘ ‘ ‘

```

Conclusion

and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.